

Scientific integrity, provenance and reproducibility audit

1. Purpose

This document separates verified evidence from interpretation. It was generated from the same machine-readable outputs used by the manuscript and report. It is not a response letter and does not upgrade a conceptual network into observed infrastructure.

2. Provenance chain

1. Public source data: LarNO rainfall, terrain and MIKE water-depth arrays.
2. Copied executable workflow: the calibrated local Itzi project was copied from `E:/Projects/20260518-itzi-flood` into `external_models/20260518-itzi-flood`; the source folder was not edited.
3. New physical products: A, B and C arrays generated locally for eight events.
4. New learning products: DrainLite models, predictions and metrics generated locally from C-B.
5. External comparator: MIKE arrays are never a DrainLite target.
6. Independent reproduction: the LarNO checkpoint metrics are calculated from a local inference array, not transcribed from Cao et al. (2026).

3. Software and fixed definitions

- Itzi 25.4; PySWMM 2.1.0; swmm-toolkit 0.17.0.
- scikit-learn 1.8.0; NumPy 2.4.5; SciPy 1.17.1; Matplotlib 3.10.9; SciencePlots 2.2.2.
- A: surface-only, Manning $n = 0.015$.
- B: surface-only, $n = 0.012$ in coupled-inlet 3 x 3 neighbourhoods.
- C: B plus native bidirectional Itzi-SWMM coupling.
- Roughness effect: B-A.
- Learning target: C-B in millimetres.
- MIKE role: external descriptive comparator only.

4. Dataset checks

The accepted dataset contains exactly eight events. Every rainfall and depth array has shape 72 x 400 x 560 and contains finite values. The static active mask contains 105,527 cells. Evaluation therefore covers 7,597,944 active cell-times per held event and 60,783,552 over the complete outer validation. The dataset audit lists no rejected event.

5. Network checks

The accepted network contains 2276 junctions, 2276 conduits and 221 receiving interfaces. Directed outfall reachability is 100.0%. Independent parsing found zero isolated junctions, cycles, duplicate undirected endpoint groups, reverse-slope conduits and crown-above-rim endpoints. SWMM ponding is disabled; surface overflow and return are handled by the 2D coupling.

6. Numerical checks

All eight events have zero SWMM flooding loss, zero warning count and zero error count. Mean routing continuity error is -0.0912% and mean non-converging-step frequency is 0.0113%. The maximum absolute combined mass error is 0.1107% of rainfall volume.

The previously observed outfall attribution count was traced to the copied reporting implementation, which repeated a global failed-step count for each outfall. Formal acceptance uses SWMM's global report value and separately captured junction counters; the repeated per-outfall field is not used in any conclusion.

7. Learning leakage checks

- Outer test units are complete events.
- Inner validation is a complete event within the outer-training set.
- Test-event targets never enter fitting, iteration selection, sample weighting or prior construction.
- For final refitting, each training event receives a prior averaged from the other six outer-training events.
- The held event receives the prior averaged from all seven outer-training events.
- Explicit row/column coordinates and distance-to-synthetic-outfall are excluded.
- Static network fields are predictors only; dynamic SWMM target-side states are excluded.

8. Quantitative claim audit

The matched surface baseline MAE to C is 9.265926 mm. The spatiotemporal prior is 3.225513 mm, prior plus dynamics is 2.183922 mm and the complete hybrid is 2.129864 mm. Static descriptors improve all eight event folds; the macro-average paired increment is 0.054058 mm with an event-bootstrap interval of 0.025694-0.082733 mm.

This evidence supports a small incremental association for the fixed network. It does not support cross-network generalisation, recovery of municipal sewer parameters, causal feature attribution, or direct application to the public LarNO checkpoint.

9. Rejected or superseded analyses

- Static depression filling is not used as a formal Itzī result.
- The early sparse 82-node/29-pipe network is superseded and excluded.
- Disconnected and high-nonconvergence SWMM runs are retained only as diagnostics.
- Sink-only labels are not used for the bidirectional model because they remove positive residuals by construction.
- The 5 m proxy is not described as 5 m MIKE validation; genuine local 5 m reference files remain unavailable.
- Previous claims based on cropped 200 x 280 windows are superseded by full 400 x 560 V3 calculations.
- No residual is added directly to the public LarNO checkpoint because its MIKE target may already include drainage.

10. File evidence

Evidence	Repository-relative path	Bytes	SHA-256
Accepted SWMM network	extended_study\output\reviewer_major_revision_v3\network\swmm_bidirectional_normal_step05.inp	510702	965978312ffa3959c21826943aa5526b90e0834b08f7698dc6455aba5f990be2
Physical quality table	extended_study\output\reviewer_major_revision_v3\formal_matched_fullphysics_quality.csv	12539	abd4736235f314b1117d04a066126e43087f0b4cbf2a7f525017183c775c40c1
Dataset audit	LarNO-main\benchmark\urbanflood\flood\region1_20m_drainage_v3_full\dataset_audit.json	464	30b08a0f252daedcb05bb4ea8d6c333f798d06d9c470233570c9028eddabaa16
Original DrainLite summary	extended_study\output\reviewer_major_revision_v3\drainlite_v3\metrics\summary_coupled.csv	1953	e5773bedc62bc8bb290fbd0903592f93c44dbe2a68b03dbec6f269f2d1bb64f9
Hybrid metadata	extended_study\output\reviewer_major_revision_v3\drainlite_hybrid_v3\experiment_metadata.json	1714	af0b5d1b2e11ce94af884b4a8c4a1be4e42ca095906dc5e7f51753cb9e8b8c79
Hybrid summary	extended_study\output\reviewer_major_revision_v3\drainlite_hybrid_v3\metrics\summary_coupled.csv	702	bcc6ebd647f2bf6b3c21a7e7087856dcdbd2c979b37b41d30cfcd75e81d9fb097
LarNO event68 prediction	LarNO-main\exp\20260220_183648_006352\pred_results\region1_20m\epoch_992\predictions_epoch_992_sample_event68.npy	64512128	b33c9b84facd93e0da1847f8139fab554be2f937b339b22334832a899e650fef

11. Figure audit

All eleven publication figures were generated from V3 CSV, JSON or NPY outputs using SciencePlots and Times New Roman. PNG files were visually reviewed at original resolution. One figure defect was found and corrected: model size in MB and process memory in GB had initially shared one numerical axis; the final Figure 11 uses separate panels and units. The workflow arrows were also repositioned after they were found to cross box labels. Map arrays use `origin=upper`, matching the audited network/DEM orientation. Percentile clipping is stated in captions and affects display only.

12. Reproduction order

1. `build_gravity_consistent_swmm_network.py`
2. `audit_reviewer_v3_network.py`
3. `run_reviewer_v3_matched_coupling.py` or `run_reviewer_v3_batch.py`
4. `merge_reviewer_v3_scenarios.py`
5. `analyze_reviewer_v3_physics.py`
6. `build_reviewer_v3_dataset.py`
7. `run_reviewer_v3_drainlite.py`
8. `run_reviewer_v3_drainlite_hybrid.py`
9. `generate_reviewer_v3_publication_figures.py`
10. `generate_reviewer_v3_documents.py`
11. `render_reviewer_v3_documents.py`

13. Remaining information before journal submission

Author list, affiliations, funding, contribution statements and data-distribution permissions are to be supplied. External observational validation, measured sewer geometry and a factorial multi-network dataset are not available. These absences are limitations, not values to be inferred or filled synthetically.